



FT43 Laboratory Pasteuriser



This self-contained laboratory unit is suitable for HTST (high temperature, short time) pasteurisation of up to 20 litres/hour of a wide variety of low-viscosity liquid foods. Products that may be pasteurised include raw milk, nectars, fruit juices, skimmed milk, ice-cream mix, soft drinks, pharmaceuticals and margarine emulsion.

PRODUCT CODE: FT43

Categories: Armfield Dairy, Armfield Educational Food Technology, Educational, FT Series - Educational Food Technology, FT43 Laboratory Pasteuriser, IFT Industrial Food Technology, IFT Series - Industrial Food Technology

Tags: CIP , Clean In Place , Data Logging , FT43 , Hold Times , HTST , Laboratory Pasteuriser , Pasteurising , Thermocouple

DESCRIPTION

This self-contained laboratory unit is suitable for HTST (high temperature, short time) pasteurisation of up to 20 litres/hour of a wide variety of low-viscosity liquid foods. Products that may be pasteurised include raw milk, nectars, fruit juices, skimmed milk, ice-cream mix, soft drinks, pharmaceuticals and margarine emulsion.

A CIP (clean-in-place) facility is incorporated which allows disinfection, pre-rinsing, detergent washing and final rinsing. The ability to vary process conditions has been built into the unit in the form of the following features:

- Variable speed, peristaltic, hygienic feed pump allowing flow rate variations
- Plate-type stainless steel heat exchanger – the cooling, regenerating and heating sections, mounted on a common frame work, are easily dismantled to allow re-configuration and inspection of the heat transfer surfaces
- Electrically heated hot water circulating system for both pasteurisation and CIP cycles – any required holding tube outlet temperature is set on the indicating microprocessor-based power output controller of the water heater
- Stainless steel holding tube, allowing holding times between five seconds and two minutes
- Flow rates of feed pump, cooling water and hot water are monitored accurately
- Temperatures in the system are monitored with thermocouple sensors at each of six chosen locations (nine available)
- Maximum process temperature 85°C

TECHNICAL SPECIFICATIONS

Optional 15-second holding tube included in supply

FEATURES & BENEFITS

BENEFITS

- Tests can be carried out quickly and easily
- Simulates the conditions used in a production plant
- Small quantities of liquid product can be processed

FEATURES

- Miniature plate heat exchanger
- Fully self-contained design with integral hot water system
- Choice of holding tubes
- Variable product temperature and holding time
- Control console incorporating comprehensive instrumentation
- CIP facility benchtop operation

Process Description

Instrumentation

- Individual conditioning for each temperature sensor
- Digital display unit with 6-way selector switch for temperature indication
- Display resolution $\pm 0.1^{\circ}\text{C}$, accuracy $\pm 1^{\circ}\text{C}$
- Data logging outputs: 6-channel analogue output 0-1V proportional to temperature
- Output port (on/off switch) for monitoring 3-way diverter-valve operation

After an initial disinfecting process, the raw liquid is placed in the feed tank [5] and pumped [4] to the regenerative section [2] of the heat exchanger. The liquid passes to the heating section [3] to be brought up to pasteurising temperature. Any product not at the required temperature after passing through the holding tube [6] is diverted back to the feed tank by a diverting valve [7] through the action of the temperature controller [10].

Heating is accomplished using water at a high flow rate only 5-6°C hotter than the pasteurising closed-circuit system [8] also controlled by temperature controller [10].

The product then passes back through the regenerator [2] and the cooler [1] to storage. (The final cooler [1] has been designed to achieve product temperatures below 10°C provided a chilled water unit is available in the laboratory.)

The regenerator is capable of achieving a high degree of heat recovery by using the surplus heat from the outgoing hot pasteurised liquid to preheat the cold liquid feed.

Process Capabilities

- Operation of a continuous HTST pasteurisation plant using selected liquid foods
- Learning the importance of Clean-In-Place (CIP) procedures
- Varying and controlling holding times
- Measuring heat transfer coefficients in a plate heat exchanger, for various plate and gasket configurations
- Measuring heat recovery in a regenerator

Accessories

FT43-DTA-ALITE Data Logging Accessory



This accessory allows the six temperature sensor outputs to be logged on a PC. The package includes all hardware, software and connecting leads required (excluding the PC itself).

The software is compatible with Windows operating systems and provides a user-friendly operator interface with a flexible range of logging and display options. The logged data can be stored and subsequently analysed by most general purpose spreadsheets.

The software requires a PC (not supplied by Armfield), running Windows 98 or above, with a USB port.

Optional 15-second holding tube included in supply

Essential Accessories and Special features

- Self-contained, bench-mounted HTST pasteuriser of up to 20 l/h capacity. CIP facility included. Main components made of stainless steel

Special features:

- Pump: hygienic design peristaltic type with variable-speed drive, maximum capacity 30 l/h (for CIP)
- Pipe fittings: stainless steel and food quality silicone tubing
- Hot water system: 1.5kW immersion type heater, self-filling water tank
- Hot water pump: magnetic-coupled circulator. Maximum pressure 0.15 bar, capacity 3 l/min
- Diverter valve: stainless steel, 3-way solenoid type
- Pasteurisation temperature controller: microprocessor-based controller, providing time-proportioned PID action to the heater, range 0-100°C, accuracy of indication $\pm 0.5^\circ\text{C}$
- Temperature measurement: 6 sheathed sensors, for use in 9 possible measurement points
- Process temperatures up to 85 °C
- A user instruction manual provides installation, commissioning and maintenance data, together with project exercises

Requirements



ELECTRICAL SUPPLY

FT43A-A: 220-240V/1ph/50Hz/13 amp

FT43A-G: 220-240V/1ph/60Hz/13 amp

Water: A supply of cold tap water is required for the cooling section. If a chilled product is required this can be achieved with a chilled water supply or by performing final chilling in a refrigerator.

Shipping Specifications

FT43A PASTEURISER

Height: 0.70m

Width: 0.80m

Depth: 0.60m

CONTROL CONSOLE

Height: 0.20m

Width: 0.35m

Depth: 0.30m

Dimensions

FT43A WITH CONSOLE

Volume: 1.0m³

Gross weight: 145kg

FT43-DTA-ALITE

Volume: 0.02m³

Gross weight: 4kg

Order Codes

FT43A-A: 220-240V/1ph/50Hz/13 amp

FT43A-G: 220-240V/1ph/60Hz/13 amp

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